

EX 7 :

Write a program to implement Logistic Regression (LR) algorithm in python

PROGRAM :

```
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
```

```
data = load_breast_cancer()
```

```
X = data.data
```

```
y = data.target
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
model = LogisticRegression(max_iter=10000)
```

```
model.fit(X_train, y_train)
```

```
y_pred = model.predict(X_test)
```

```
print("Predicted Values:")
```

```
print(y_pred)
```

```
print("\nAccuracy:")
```

```
print(accuracy_score(y_test, y_pred))
```

OUTPUT

```
... Predicted Values:  
[1 0 0 1 1 0 0 0 1 1 1 0 1 0 1 0 1 1 1 0 1 1 0 1 1 1 1 1 1 0 1 1 1 1 1 0  
 1 0 1 1 0 1 1 1 1 1 1 1 1 0 0 1 1 1 1 1 0 1 1 1 0 0 1 1 1 0 0 1 1 0 0 1 0  
 1 1 1 1 1 1 0 1 1 0 0 0 0 0 0 1 1 1 1 1 1 1 0 0 1 0 0 1 0 0 1 1 1 0 1 1 0  
 1 0 0]  
  
Accuracy:  
0.956140350877193
```